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Carbon nanotubes modified graphene hybrid Aerogel-based composite phase change materials for efficient thermal storage

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ABSTRACT

Phase change materials (PCMs) have attracted more and more attention in the field of latent heat storage. In order to solve the leakage behavior and improve heat transfer performance of polyethylene glycol (PEG), a method of self-assembled hybrid aerogel filled with carbon nanotubes (CNTs) combined with reduced graphene oxide (rGO) nanosheets is proposed herein. The as-prepared CNTs-graphene hybrid aerogel (CGA) with three-dimensional porous structure can be used as a highly adsorptive and stable carrier for PCM. After the vacuum impregnation process, a high-performance PEG/CGA composite PCMs (c-PCMs) is obtained, which exhibits an outstanding PEG loading ratio of 97.6 %. Moreover, after 1000 repeated heating/cooling cycles, the H_M and H_S value of PEG/CGA c-PCMs remained 93.44 % and 92.84 %. Meanwhile the PEG/CGA c-PCMs remains intact and the loading ratio could still maintain as high as 96.4 %, which proved that PEG/CGA c-PCMs presented reliable thermal cycling stability. The PEG/CGA c-PCMs only takes 603 s to raise from 30 ℃ to 70 ℃, which is much faster than that of pure PEG (813 s) and PEG/GA composite (727 s). And The thermal conductivity of PEG/CGA(0.586 W/(m·K)) was 186 % times than that of pure PEG(0.314 W/(m·K)).The introduction of CNTs can not only enhance the adsorption capacity of c-PCMs and alleviate the leakage of PEG, but also improve the heat transfer performance of c-PCMs. These results specify that PEG/CGA c-PCMs can be used as remarkable and stable energy storage materials for environmental thermal management systems.

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1. Introduction

As various countries and regions around the world have proposed "carbon neutrality" or "zero carbon" goals, "green and low carbon" and "sustainable development" have become an international consensus [1]. This has fostered the development of sustainable and renewable energies, among which solar energy is widely regarded as the most important renewable energy source [2]. But the potential of solar energy is limited, mainly due to its intermittency, regional variability and unpredictability [3,4]. The unique properties of phase change materials (PCMs) were precisely solved this limitation, which could store a large amount of thermal energy through the phase change process and be used as a suitable medium for solar thermal energy conversion and storage [5,6]. Currently, PCMs including inorganic and organic PCMs, which have been used in many fields such as building energy conservation, thermal insulation, smart textiles, waste heat recovery and solar heating systems [7–9]. Compared with inorganic PCMs, affordable organic PCMs offer the benefits of moderate phase change enthalpy, wide melting temperature, and well-behaved chemical and thermal stability, which have been extensively investigated [7,10,11]. Among the various organic PCMs, polyethylene glycol (PEG) presents the advantages of high latent heat of phase change, low cost, non-toxicity, well-behaved chemical and thermal stability, etc., which makes it a promising solid–liquid organic PCM. However, the main imperfection such as the leak problem and the poor heat transfer during thermal storage and release process still limit its further application [12–16].

To tackle these two problems, some researchers have concentrated on improving the shape stability and heat transfer performance of PCMs. For solving the leak problem, the common method is to encapsulate PCMs into the matrix material to construct a shape-stabilized PCMs (ss-PCMs). The matrix materials with large specific surface area and excellent stability can control the fluidity of organic PCMs through physical interaction, thereby preventing the leak problem of PCMs [17–20]. For example, graphene aerogel (GA) [3,21–24], metal foam [25–30], porous carbon [12,13,31,32] and porous mineral materials [14,15,33,34] have

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